

Linear Algebra

MAT 226

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Homework 4

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Exercise 1.7

$$2) \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -8 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 2 & 0 & 3 & 0 \\ 3 & -8 & 1 & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 2 & 0 & 3 & 0 \\ 0 & 0 & -1 & 0 \\ 3 & -8 & 1 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 2 & 0 & 3 & 0 \\ 3 & -8 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - \frac{3}{2} R_1$$

$$\begin{bmatrix} 2 & 0 & 3 & 0 \\ 0 & -8 & -3.5 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

The matrix is in echelon form and has a pivot in every column. There are no free variables, hence the equations have only the trivial solution.

The vectors are linearly independent.

$$4) \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ -9 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -3 & 0 \\ 3 & -9 & 0 \end{bmatrix}$$

$$R_1 \leftarrow -R_1$$

$$R_2 \leftarrow R_2 / 3$$

$$\begin{bmatrix} 1 & 3 & 0 \\ 1 & -3 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & -6 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / 6$$

$$\begin{bmatrix} \textcircled{1} & 3 & 0 \\ 0 & \textcircled{1} & 0 \end{bmatrix}$$

The matrix is in echelon form and has a pivot in every column. There are no free variables, hence the equations have only the trivial solution.

The vectors are linearly independent.

Shorter method

$$\text{let } v_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} -3 \\ -9 \end{bmatrix}$$

v_1 & v_2 would be linearly dependent if & only if one is a scalar multiple of the other

$$3v_1 = v_2$$

$$3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 9 \end{bmatrix} \neq \begin{bmatrix} -3 \\ -9 \end{bmatrix}$$

hence v_1 & v_2 are linearly independent.

$$6) \begin{bmatrix} -4 & -3 & 0 \\ 0 & -1 & 5 \\ 1 & 1 & -5 \\ 2 & 1 & -10 \end{bmatrix} \Rightarrow \begin{bmatrix} -4 & -3 & 0 & 0 \\ 0 & -1 & 5 & 0 \\ 1 & 1 & -5 & 0 \\ 2 & 1 & -10 & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & -5 & 0 \\ 0 & -1 & 5 & 0 \\ -4 & -3 & 0 & 0 \\ 2 & 1 & -10 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 + 4R_1$$

$$R_4 \leftarrow R_4 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & -5 & 0 \\ 0 & -1 & 5 & 0 \\ 0 & 1 & -20 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 + R_2$$

$$R_4 \leftarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & 1 & -5 & 0 \\ 0 & -1 & 5 & 0 \\ 0 & 0 & -15 & 0 \\ 0 & 0 & -5 & 0 \end{bmatrix}$$

Every variable column contains atleast one pivot. There are no free variables hence the columns of the matrix are linearly independent.

$$8) \begin{bmatrix} 1 & -2 & 3 & 2 \\ -2 & 4 & -6 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -2 & 3 & 2 & 0 \\ -2 & 4 & -6 & 2 & 0 \\ 0 & 1 & -1 & 3 & 0 \end{bmatrix}$$

There are 4 variable columns & only 3 rows. Atleast one variable must be free hence trivial solution is not the only solution. The columns of the matrix do not form a linear independent set.

$$12) \begin{bmatrix} 3 \\ -6 \\ 1 \end{bmatrix}, \begin{bmatrix} -6 \\ 4 \\ -3 \end{bmatrix}, \begin{bmatrix} 9 \\ h \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -6 & 9 & 0 \\ -6 & 4 & h & 0 \\ 1 & -3 & 3 & 0 \end{bmatrix}$$

$$R_3 \leftrightarrow R_3 \div 3$$

$$\begin{bmatrix} 1 & -2 & 3 & 0 \\ -6 & 4 & h & 0 \\ 1 & -3 & 3 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 + 6R_1$$

$$R_3 \leftarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & -2 & 3 & 0 \\ 0 & -8 & h+18 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 2R_3$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & -8 & h+18 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & -8 & h+18 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 8R_2$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & h+18 & 0 \end{bmatrix}$$

for the vectors to be linearly dependent, there needs to be a free variable which is only possible when $h+18=0$
 $h = -18$

vectors are linearly independent when $h = -18$

2.2) True

(a) u, v are linearly independent
 w is in $\text{span}(u, v)$
it means $w = c_1 u + c_2 v$

then

$$c_1 u + c_2 v + c_3 (c_1 u + c_2 v)$$

$$c_1 u + c_2 v + c_1 c_3 u + c_2 c_3 v$$

$$u(c_1 + c_1 c_3) + v(c_2 + c_2 c_3) \quad (\text{the scalars change})$$

This would also be linearly independent

Hence, statement is true.

(b) True

(e.g.) $v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} 3 \\ 5 \\ 0 \end{bmatrix}$ $v_3 = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$

In augmented matrix form, there would be a free variable hence the vectors would be linearly dependent.
hence True

(c) False

it means more rows than columns

but (e.g.) if a vector is multiple of the other

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \text{ they are linearly dependent}$$

(d) False

A set can be linearly dependent with n or fewer vectors.

e.g. if it has a $\vec{0}$ vector.

34) v_1 and v_2 in $\mathbb{R}^4$ (not multiple)

$$(e.g) \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

v_1 & v_2 are linearly dependent

so the statement is false.

Exercise 1.8

$$2) \quad A = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \quad u = \begin{bmatrix} 3 \\ 6 \\ -9 \end{bmatrix} \quad v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$T(x) = Ax$$

$$T(u) = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ -9 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{3} \times 3 + 0 \times 6 + 0 \times -9 \\ 0 \times 3 + \frac{1}{3} \times 6 + 0 \times -9 \\ 0 \times 3 + 0 \times 6 + \frac{1}{3} \times -9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

$$T(v) = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a \times \frac{1}{3} + 0 + 0 \\ 0 + b \times \frac{1}{3} + 0 \\ 0 + 0 + c \times \frac{1}{3} \end{bmatrix} = \begin{bmatrix} a/3 \\ b/3 \\ c/3 \end{bmatrix}$$

$$4) \quad A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -3 \\ 2 & -5 & 6 \end{bmatrix} \quad b = \begin{bmatrix} -6 \\ -4 \\ -5 \end{bmatrix}$$

$$\text{Augmented matrix } [A|b] = \begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 2 & -5 & 6 & -5 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 0 & -1 & 0 & 7 \end{bmatrix}$$

$$R_3 \leftarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & -3 & 3 \end{bmatrix}$$

$$R_3 \leftarrow R_3 \times -\frac{1}{3}$$

$$\begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_3 \quad R_2 \leftarrow R_2 + 3R_3$$

$$\begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$R_1 \leftarrow R_1 + 2R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & -17 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$x_1 = 17$$

$$x_2 = 7$$

$$x_3 = 1$$

$$x = \begin{bmatrix} 17 \\ 7 \\ 1 \end{bmatrix}$$

x is unique.

$$10) \quad A = \begin{bmatrix} 3 & 2 & 10 & -6 \\ 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 1 & 4 & 10 & 8 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3 \quad R_1 \leftrightarrow R_4$$

$$[A|b] = \begin{bmatrix} 1 & 4 & 10 & 8 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 1 & 0 & 2 & -4 & 0 \\ 3 & 2 & 10 & -6 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_1$$

$$R_4 \leftarrow R_4 - 3R_1$$

$$\begin{bmatrix} 1 & 4 & 10 & 8 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & -12 & 0 \\ 0 & -10 & -20 & -30 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 10 & 8 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 3 & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_2$$

$$R_4 \leftarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & 4 & 10 & 8 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 4R_2$$

$$\begin{bmatrix} 1 & 0 & 2 & -4 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + 2x_3 - 4x_4 = 0$$

$$x_2 + 2x_3 + 3x_4 = 0$$

x_3, x_4 are free variables.

all x in $\mathbb{R}^4$

$$x = \begin{bmatrix} -2x_3 + 4x_4 \\ -2x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 4 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

12)

$$A = \begin{bmatrix} 3 & 2 & 10 & -6 \\ 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 1 & 4 & 10 & 8 \end{bmatrix}$$

$$b = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 10 & -6 & -1 \\ 1 & 0 & 2 & -4 & 3 \\ 0 & 1 & 2 & 3 & -1 \\ 1 & 4 & 10 & 8 & 4 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$R_1 \leftrightarrow R_4$$

$$\begin{bmatrix} 1 & 4 & 10 & 8 & 4 \\ 0 & 1 & 2 & 3 & -1 \\ 1 & 0 & 2 & -4 & 3 \\ 3 & 2 & 10 & -6 & -1 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_1$$

$$R_4 \leftarrow R_4 - 3R_1$$

$$\begin{bmatrix} 1 & 4 & 10 & 8 & 4 \\ 0 & 1 & 2 & 3 & -1 \\ 0 & -4 & -8 & -12 & -1 \\ 0 & -10 & -20 & -30 & -10 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 4 & 10 & 8 & 4 \\ 0 & 1 & 2 & 3 & -1 \\ 0 & 4 & 8 & 12 & -1 \\ 0 & 1 & 2 & 3 & 1 \end{bmatrix}$$

$$R_4 \leftarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & 4 & 10 & 8 & 4 \\ 0 & 1 & 2 & 3 & -1 \\ 0 & 4 & 8 & 12 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

Row 4 means $0x_1 + 0x_2 + 0x_3 + 0x_4 = 2$

the system is inconsistent

There is no possible vector x in $\mathbb{R}^4$ that can be multiplied by A to produce b .

$$20) \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; \quad v_1 = \begin{bmatrix} -3 \\ 5 \end{bmatrix} \quad v_2 = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$$

$$T(x) = x_1 v_1 + x_2 v_2 = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A x$$

$$A = \begin{bmatrix} -3 & 7 \\ 5 & -2 \end{bmatrix}$$

22) (a) True

because the range is the set of all possible values.
(all possible Ax vectors).

(b) True

for a transformation to be linear, it must preserve vector addition & scalar multiplication.

$$\text{matrixes obey } A(u+v) = Au + Av$$

$$\& \quad A(cu) = c(Au) = u(Ac)$$

(c) It asks if there is at least one x that maps to c via transformation whereas uniqueness would mean checking if there is only one c or many.

False

d) True

Shown in (b)

$$A(u+v) = Au + Av$$

$$A(cu) = c(Au) = u(Ac)$$

e) True

The zero vector in domain always maps to zero vector in co-domain

$$T(0) = T(0 \cdot x) = 0 \cdot T(x) = 0$$

$$24) \quad T(x) = Ax + b$$

test with $\vec{0}$

$$T(\vec{0}) = A(\vec{0}) + b$$

$$T(0) = 0 + b$$

$$T(0) = b$$

since $b \neq 0$ hence $T(0) \neq 0$

It does not map origin to origin hence it is not a linear transformation.

$$\begin{aligned} \text{or} \\ T(u+v) &= A(u+v) + b \\ &= A(u) + A(v) + b \end{aligned}$$

$$\begin{aligned} T(u+v) \text{ should be } & T(u) + T(v) \\ & T(u) + T(v) \\ & A(u) + b + A(v) + b \\ & A(u) + A(v) + 2b \end{aligned}$$

$$A(u) + A(v) + b \neq A(u) + A(v) + 2b \quad (b \neq 0)$$

hence Transformation not linear